

Integrated Geneformer Analysis of Lung Cancer Single-Cell Transcriptomes

T-cell and Epithelial Cell Analysis in Normal Lung, Adenocarcinoma, and Small Cell Lung Cancer

Kaisar Dauvey, Minh-Huyen Hoang

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Short abstract Geneformer identified preliminary disease-specific transcriptional programs in epithelial tumor cells, with LUAD versus SCLC epithelial classification achieving validation accuracy of 0.951 and macro-F1 of 0.941. In contrast, T-cell transcriptomes showed moderate disease separation, suggesting that immune cells capture shared tumor-associated immune remodeling rather than tumor lineage alone. Perturbation analysis nominated candidate genes associated with LUAD-to-SCLC transcriptional transition.

1 Study Overview

We applied Geneformer to single-cell RNA sequencing data from normal lung, lung adenocarcinoma (LUAD), and small cell lung carcinoma (SCLC) published by Chan et al. 2021 (<https://cellxgene.cziscience.com/collections/62e8f058-9c37-48bc-9200-e767f318a8ec>). The analysis focused on two biologically complementary compartments:

1. **T cells**, representing the immune microenvironment.
2. **Epithelial cells**, representing the tumor-lineage compartment.

The full dataset contained approximately 147,137 cells and 22,021 genes.

Table 1: Disease composition of the full dataset

Disease group	Number of cells
Small cell lung carcinoma	77,143
Lung adenocarcinoma	57,679
Normal lung	12,315

2 T-cell Geneformer Analysis

2.1 Objective

The T-cell analysis tested whether disease-associated immune states can distinguish normal lung, LUAD, and SCLC from T-cell transcriptomes alone.

Table 2: T-cell disease-state classifier performance

Metric	Value
Accuracy	0.661
Macro-F1	0.499

2.2 Classifier Performance

2.3 Interpretation

The T-cell classifier detected disease-associated signal, but separation was incomplete. LUAD-associated T-cell states were most readily identified, while SCLC and normal T cells were frequently misclassified as LUAD. This suggests substantial overlap between tumor-associated immune programs in LUAD and SCLC.

Clinical implication: T-cell transcriptomes reflect immune remodeling and may be useful for studying immune checkpoint biology, but they are less reliable for determining tumor lineage than epithelial tumor cells.

3 Epithelial Geneformer Analysis

3.1 Objective

The epithelial analysis tested whether tumor epithelial transcriptomes can distinguish LUAD from SCLC and whether perturbation analysis can nominate genes associated with LUAD-to-SCLC-like transition.

3.2 Dataset

The main epithelial subset contained 31,258 epithelial cells.

Table 3: Main epithelial subset used for LUAD versus SCLC classification

Disease group	Epithelial cells	Donors
Lung adenocarcinoma	5,270	16
Small cell lung carcinoma	25,988	9

3.3 Classifier Performance

Table 4: LUAD versus SCLC epithelial classifier performance

Metric	Value
Validation accuracy	0.951
Validation macro-F1	0.941
Validation loss	0.221
Training loss	0.0395

3.4 Interpretation

The epithelial classifier strongly separated LUAD and SCLC, indicating that epithelial tumor cells carry robust lineage-specific transcriptional information. This result is biologically plausible

because LUAD and SCLC differ substantially in differentiation state, neuroendocrine identity, and tumor lineage programs.

4 Exploratory Three-Class Epithelial Classifier

A second epithelial classifier included normal lung epithelial cells together with LUAD and SCLC.

Table 5: Exploratory three-class epithelial classifier performance

Metric	Value
Validation accuracy	0.850
Validation macro-F1	0.747

This result should be interpreted as exploratory because the normal group was smaller and the notebook suggested possible donor overlap in the three-class split. A strict donor-exclusive split should be used in future runs.

5 In Silico Epithelial Perturbation

5.1 Objective

The perturbation analysis modeled a disease-state transition:

LUAD epithelial state $\rightarrow$ SCLC epithelial state

5.2 Perturbation Design

- Perturbation type: *in silico* overexpression.
- Genes perturbed: all detected genes.
- Model: fine-tuned LUAD/SCLC epithelial Geneformer classifier.
- Maximum cells used: 2,500.

5.3 Perturbation Results

The perturbation screen evaluated 15,531 genes. After filtering for false discovery rate and detection frequency, 1,059 genes remained significant.

These genes are computational candidates and require orthogonal validation. Broadly expressed genes such as *ACTB*, *TPT1*, and *PTMA* should be interpreted cautiously.

6 Integrated Cross-Compartment Interpretation

The combined analyses support the following model:

Tumor epithelial cells encode primary LUAD/SCLC lineage identity.
T cells reflect downstream immune remodeling and shared tumor-associated states.

Table 6: Representative perturbation candidates

Direction	Example genes	Interpretation
Positive shift toward SCLC	<i>SFTPC, TPT1, SLPI, PTMA, ACTB</i>	Candidate LUAD-to-SCLC-like transition genes
Negative shift away from SCLC	<i>B2M, S100A9, FTH1, RALGAPA2, ITGB8</i>	Genes moving cells away from SCLC-like state

Table 7: Comparison of T-cell and epithelial Geneformer analyses

Feature	T cells	Epithelial cells
Disease classification	Moderate	Strong
LUAD vs SCLC separation	Limited	Excellent
Primary biological signal	Immune state	Tumor lineage
Clinical relevance	Immunotherapy biology	Tumor identity and lineage plasticity

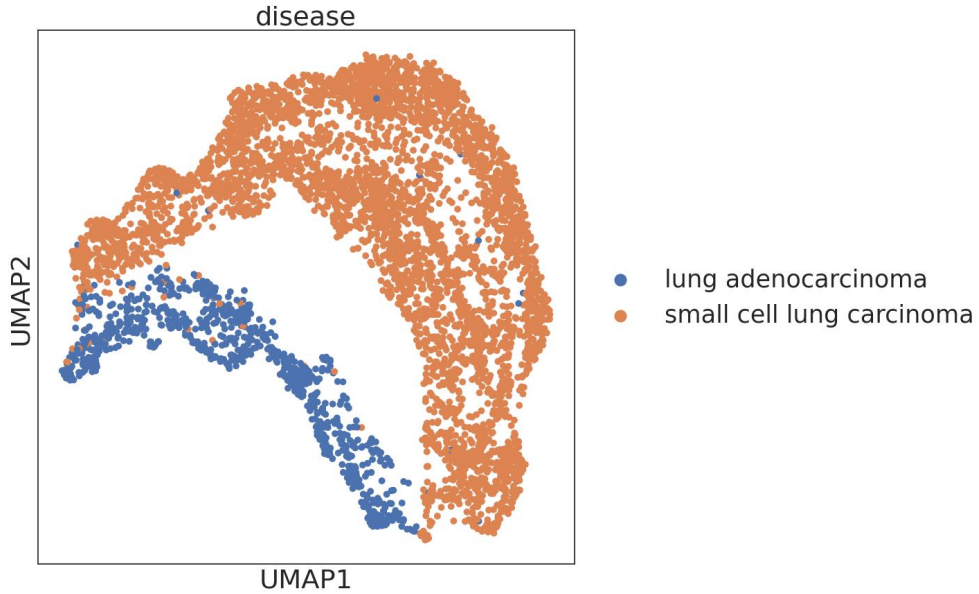


Figure 1: Geneformer epithelial embeddings showing disease-associated separation between LUAD and SCLC epithelial cells.

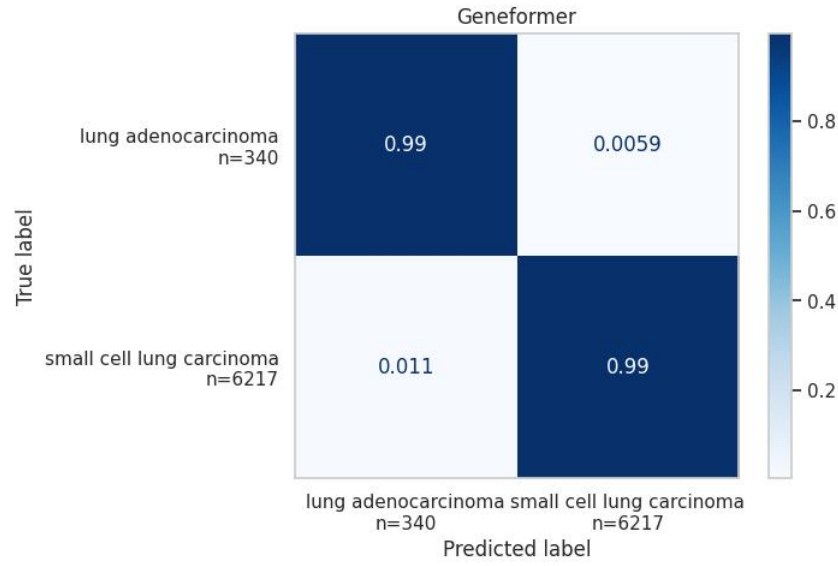


Figure 2: Confusion matrix for LUAD versus SCLC epithelial Geneformer classifier. The validation accuracy was 0.951 and macro-F1 was 0.941.

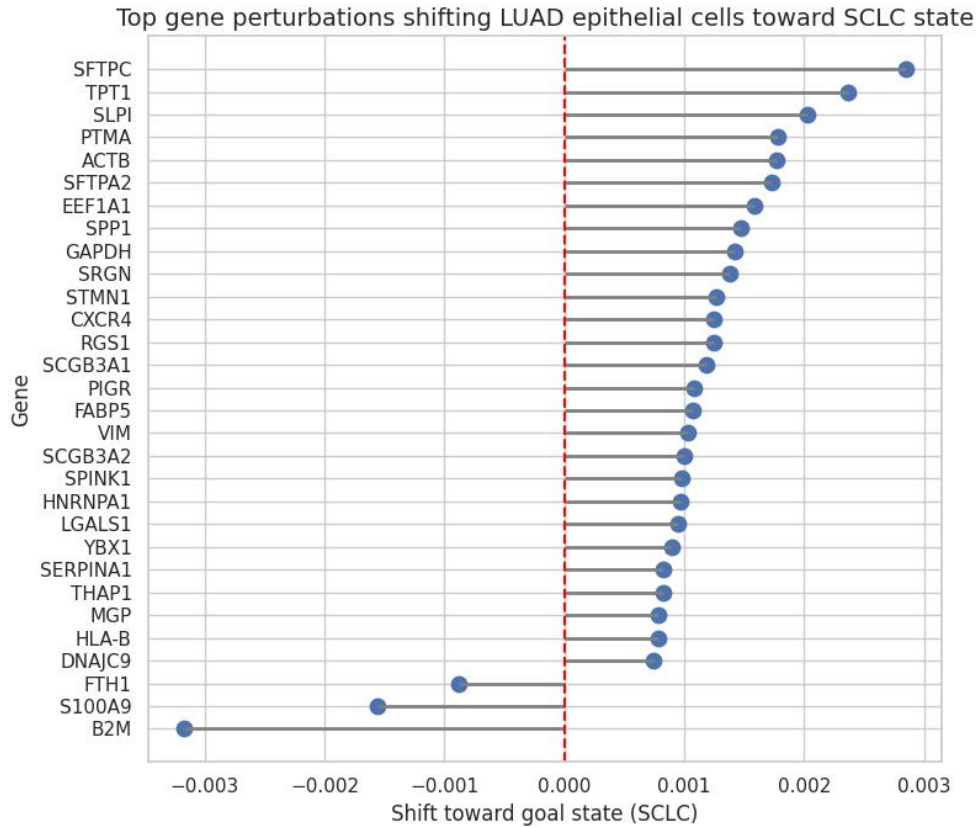


Figure 3: Geneformer perturbation analysis identifying genes whose simulated overexpression shifts LUAD epithelial cells toward an SCLC-like state.

7 Limitations

- Epithelial LUAD/SCLC analysis was imbalanced by cell number and donor number.
- Geneformer V1 was used in the epithelial notebook; V2-104M should be tested.
- The three-class epithelial analysis may have donor leakage and should be rerun with strict donor-exclusive splitting.
- Perturbation outputs are computational predictions, not experimental evidence.
- Highly expressed housekeeping genes require careful biological filtering.

8 Planned Next Steps

1. Re-run T-cell and epithelial analyses with large size non-overlapping train/eval/test splits.
2. Compare Geneformer V1 and V2-104M.
3. Add external normal lung T-cell references such as HLCA.
4. Prioritize epithelial lineage genes: *ASCL1*, *NEUROD1*, *POU2F3*, *YAP1*, *MYC*, and NOTCH pathway genes.
5. Prioritize immune checkpoint and exhaustion genes: *PDCD1*, *CTLA4*, *TIGIT*, *LAG3*, *IFNG*, *TNF*, *CXCL13*, and *TOX*.
6. Perform pathway enrichment on positive and negative perturbation hits separately.
7. Validate top candidates using independent datasets and in silico perturbation experiments from in vivo published studies.

9 Summary

Geneformer was used to analyze T-cell and epithelial compartments from lung cancer single-cell RNA-seq data. T-cell classification across normal lung, LUAD, and SCLC showed moderate performance, suggesting that immune cells capture disease-associated but overlapping cancer immune states. In contrast, epithelial tumor cells showed strong LUAD-versus-SCLC separation, with validation accuracy of 0.951 and macro-F1 of 0.941. This indicates that epithelial cells carry the strongest tumor-lineage signal. In silico perturbation further nominated candidate genes associated with LUAD-to-SCLC-like transition. These findings support Geneformer as a useful framework for studying tumor identity, immune remodeling, and computational perturbation in lung cancer.